LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 4 Test (GSEB pattern): Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds

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SECTION A

Q1. Ethane, with the molecular formula C_2H_6, has: **[1]**

Answer: (b) 7 covalent bonds: one C–C bond and six C–H bonds.

Q2. The next homologue of methane (CH_4) is: **[1]**

Answer: (a) C_2H_6 (ethane)

Q3. Butanone is a four-carbon compound with the functional group: **[1]**

Answer: (c) ketone

Q4. The functional group present in propanal is: **[1]**

Answer: (d) aldehyde (–CHO)

Q5. The next higher homologue of ethanol (C_2H_5OH) is: **[1]**

Answer: (b) C_3H_7OH (propanol). Consecutive members differ by $–CH_2–$.

Q6. The two carbon atoms in ethene are joined by: **[1]**

Answer: (c) a double bond

Q7. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that: **[1]**

Answer: (a) the fuel is not burning completely (the soot is unburnt carbon).

Q8. Soaps are: **[1]**

Answer: (d) sodium or potassium salts of long-chain carboxylic acids

Q9. Which of the following is an unsaturated hydrocarbon? **[1]**

Answer: (b) C_2H_4 (ethene). It has a carbon–carbon double bond.

SECTION B

Q10. What is a homologous series? Write the formulae of the first two members of the alkyne series. **[2]**

Answer: A series of compounds in which the same functional group replaces hydrogen in a carbon chain, and consecutive members differ by a $–CH_2–$ unit (14 u). Alkynes: ethyne C_2H_2 and propyne C_3H_4.

- Definition: 1
- Two formulae: ½ + ½

Q11. Why do covalent compounds generally have low melting and boiling points? Why are they poor conductors of electricity? **[2]**

Answer: The forces between their molecules (intermolecular forces) are weak, so little energy is needed to separate them. They are poor conductors because the bonds are formed by sharing electrons, so no ions are formed.

- Weak intermolecular forces: 1
- No ions: 1

Q12. What is saponification? Write the balanced equation for the reaction of ethyl ethanoate with sodium hydroxide. **[2]**

Answer: The reaction of an ester with an alkali (sodium hydroxide) that gives back the alcohol and the sodium salt of the carboxylic acid. It is used to make soap. $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$

- Definition: 1
- Balanced equation: 1

SECTION C

Q13. Answer the following about ethanol: **[3]**

(i) $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$. Hydrogen gas is given off and sodium ethoxide is formed. [1]
(ii) Ethene is formed (dehydration): $C_2H_5OH \rightarrow C_2H_4 + H_2O$ [1]
(iii) Ethanoic acid (CH_3COOH) [1]

Q14. Answer the following: **[3]**

(i) C_nH_{2n} [1]
(ii) Propane, C_3H_8 [1]
(iii) 14 u (one $–CH_2–$ unit) [1]

SECTION D

Q15. Carbon compounds burn in oxygen to give carbon dioxide and water, along with heat and light. Saturated hydrocarbons generally burn with a clean blue… **[4]**

(i) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light [1]
(ii) Unsaturated hydrocarbons. [1]
(iii) There is not enough oxygen, so combustion is incomplete and unburnt carbon (soot) is formed. This wastes fuel, blackens vessels, and incomplete combustion can also produce poisonous carbon monoxide. [2]

- Incomplete combustion due to less oxygen: 1
- One problem: 1